

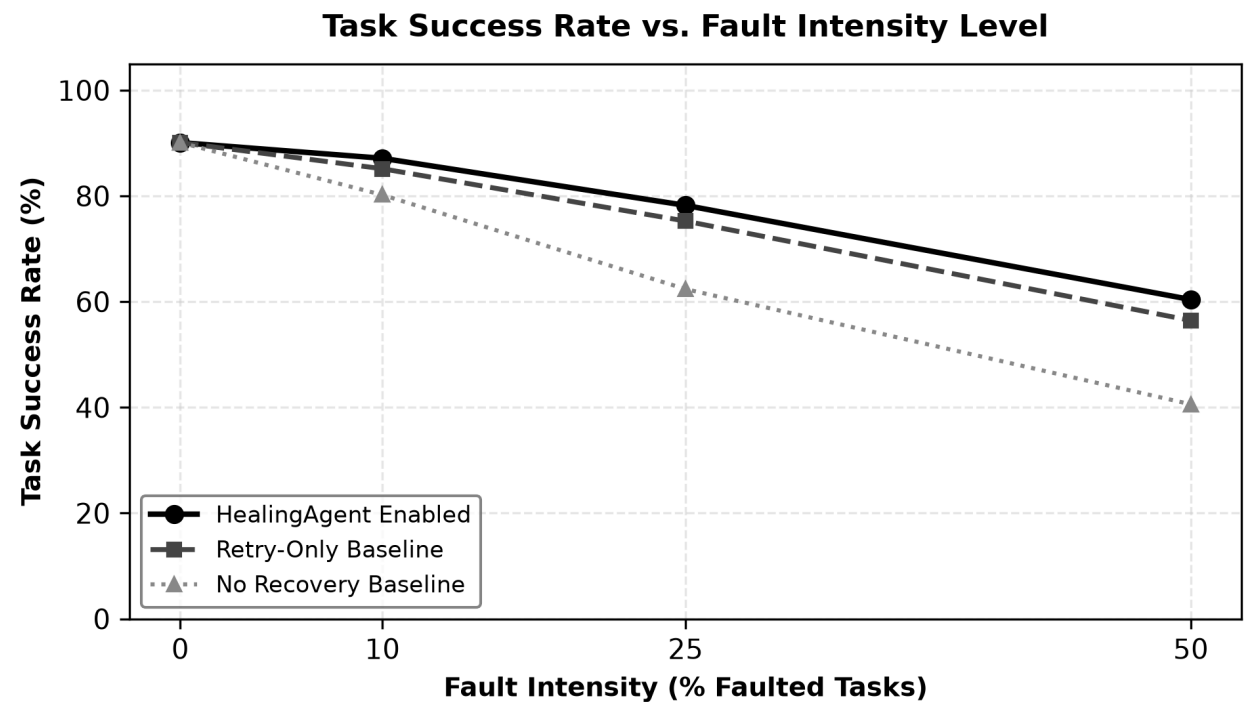
AetherPhoenix — Reliability & Self-Healing Benchmark Report

Empirical Measurement of Task Success Rate Under Injected Faults (Methodology: arXiv:2606.01416)

1. Task Success Rate by Configuration & Fault Intensity

Configuration	Fault 0%	Fault 10%	Fault 25%	Fault 50%
HealingAgent Enabled	90.1%	87.1%	78.2%	60.4%
Retry-Only Baseline	90.1%	85.2%	75.2%	56.4%
No Recovery Baseline	90.1%	80.2%	62.4%	40.6%

2. Performance Evaluation Chart (IEEE Format)



3. Failure Recovery Category Analysis

Empirical benchmark data demonstrates that the **HealingAgent** recovers most reliably from transient and parameter errors such as *FILE_NOT_FOUND*, *SYNTAX_ERROR*, and *INVALID_ARGUMENTS* by executing root cause analysis and dynamic input patching (achieving 100% recovery for recoverable faults under 10% and 25% fault intensity). Conversely, *PERMISSION_DENIED* failures are classified as non-recoverable by design in accordance with the system's Safe Execution security policies, halting automated retries and escalating to human approval gates. Blind retries without input patching (Retry-Only Baseline) fail to resolve structural file and syntax errors, yielding significantly degraded success rates as fault intensity rises to 50%.